Hash tables exercises

11-E01 Load factor and bucket events

Difficulty: Basic

Let m=12,n=18, with keys independently and uniformly assigned to chaining buckets. Find the load factor and expected unsuccessful-search cost. Distinguish the probability that all keys enter one specified bucket from the probability that all share some bucket.

11-E02 Modulo collisions at m=2^p

Difficulty: Advanced

For h(k)=k mod 8, give at least two pairs of distinct keys with equal low bits. Explain why they always collide.

11-E03 Deletion in open addressing

Difficulty: Advanced

Let m=7 and h(k)=k mod 7. Insert 1 and 8; linear probing places 8 in slot 2 after slot 1. What happens if 1 is simply erased and 8 is searched for? Give a correct treatment.

11-E04 Exact finite universal-family probability

Difficulty: Research

Let P=5,m=3, a uniform on 1..4, and b uniform on 0..4. For fixed x=0,y=1, compute the exact collision probability of h_(a,b)(k)=((ak+b) mod 5) mod 3 and compare it with 1/3.

11-E05 Double-hashing coverage period

Difficulty: Advanced

Let m=12, home=1, step=8. List probes until the first return home and count reachable slots. Why does step=5 cover the whole table? Can the first scheme claim that the table is full when an unreachable slot is empty?

11-E06 Perfect hashing still compares original keys

Difficulty: Research

Use K={4,9,15,20,26,31}, P=37, first-level h=k mod 37 mod 6, and second-level h_j=k mod 37 mod 4 for the two-key buckets. List buckets and total second-level slots; trace queries 26 and 14. Why does collision freedom not eliminate the final equality check?

Hash tables exercises (continued)

> This file contains questions only. Complete your derivations before opening the separate solutions PDF.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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